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(54) **SYSTEMS AND METHODS FOR
SUBSCRIPTION BASED SELECTIVE EVPN
ROUTE DISTRIBUTION**

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

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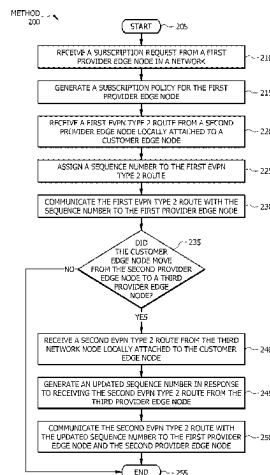
(51) **Int. Cl.**
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CPC **H04L 12/4641** (2013.01); **H04L 45/02**
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In one embodiment, a method includes receiving, by a route
reflector, a subscription request from a first provider edge
node in a network and generating a subscription policy for
the first provider edge node. The method also includes
receiving a first Ethernet Virtual Private Network (EVPN)
Type 2 Route from a second provider edge node, assigning
a sequence number to the first EVPN Type 2 Route, and
communicating the first EVPN Type 2 Route with the
sequence number to the first provider edge node. The
method further includes receiving a second EVPN Type 2
Route from a third provider edge node, generating an
updated sequence number in response to receiving the
second EVPN Type 2 Route from the third provider edge
node, and communicating the second EVPN Type 2 Route
with the updated sequence number to the first provider edge
node and the second provider node.

24 Claims, 3 Drawing Sheets



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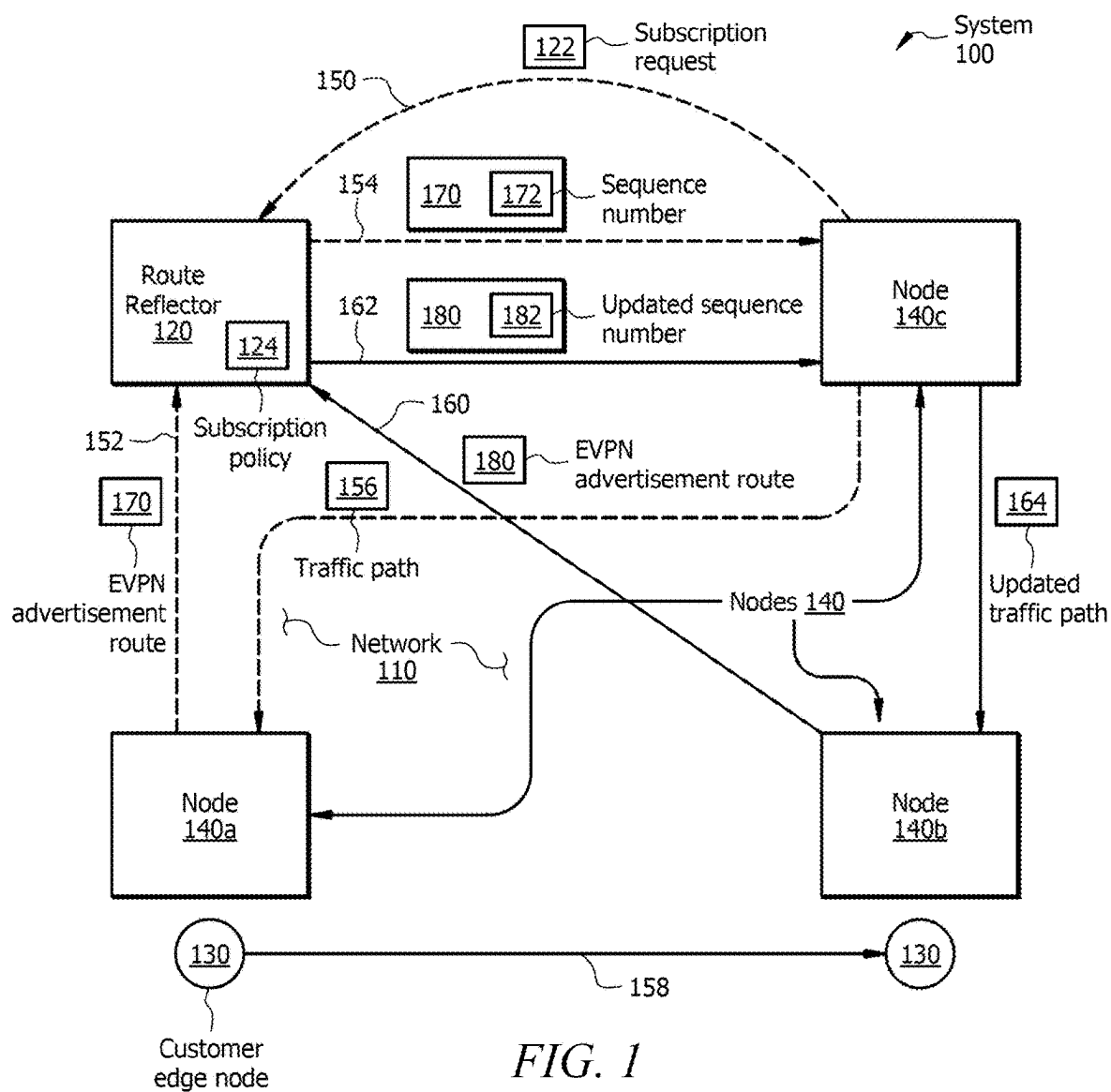
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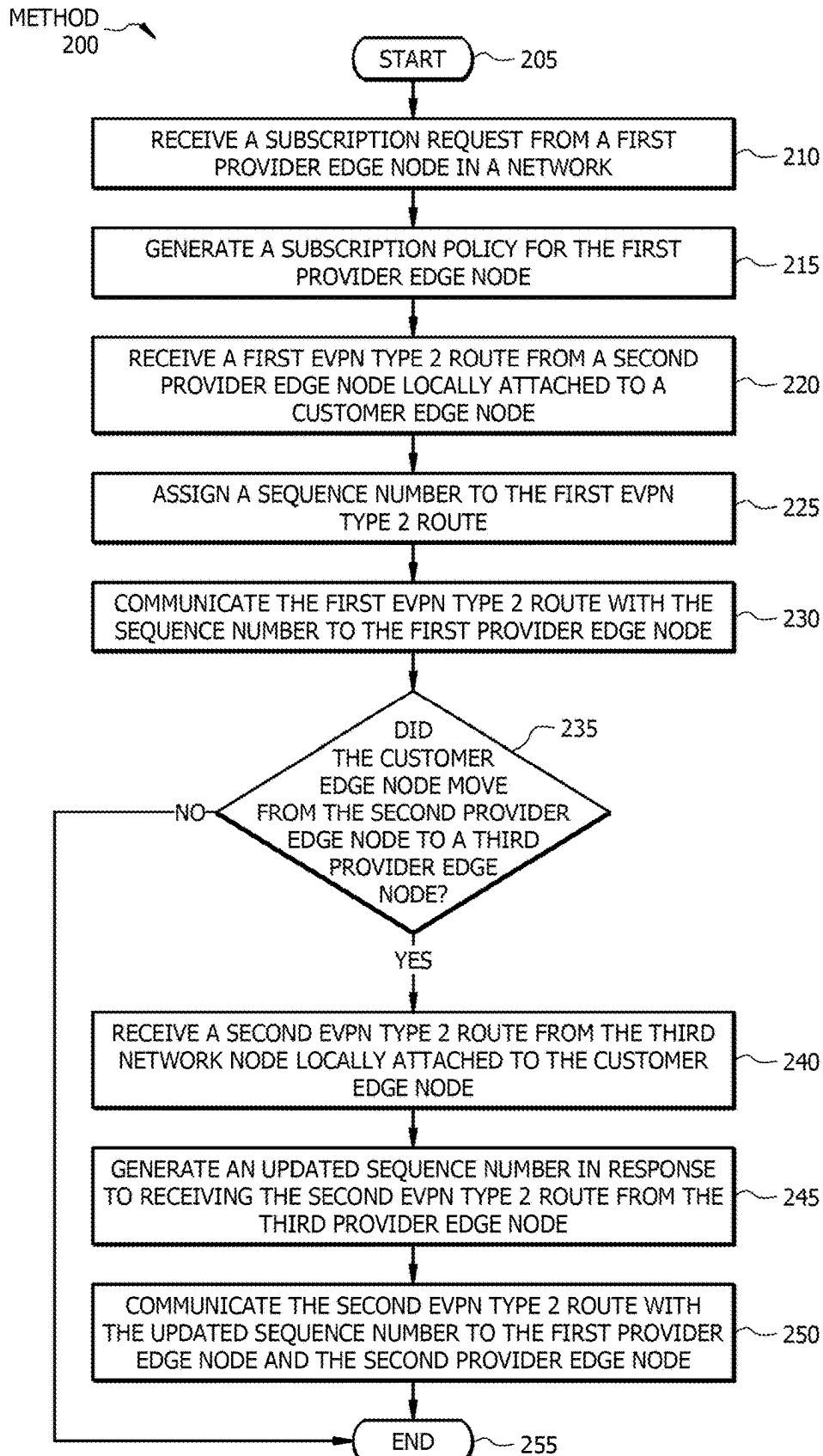
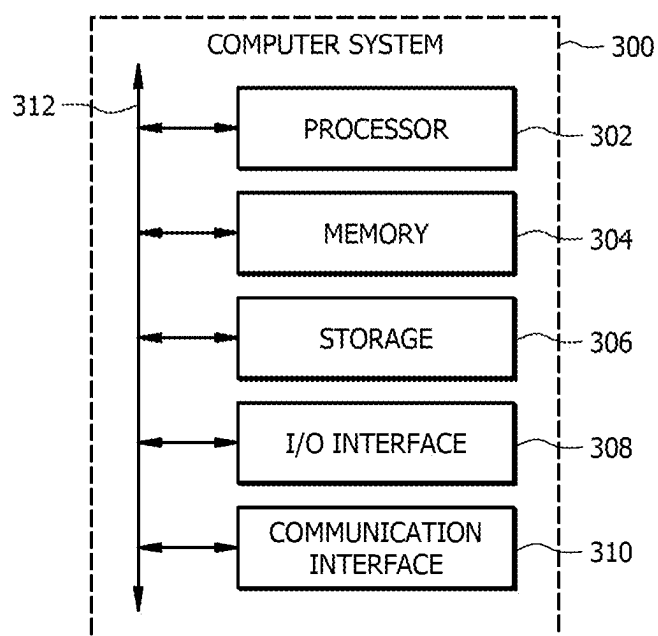


FIG. 2

*FIG. 3*

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SYSTEMS AND METHODS FOR SUBSCRIPTION BASED SELECTIVE EVPN ROUTE DISTRIBUTION

PRIORITY

This nonprovisional application is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 18/641,898 filed on Apr. 22, 2024, which is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 17/813,959 filed on Jul. 21, 2022, now U.S. Pat. No. 11,979,256, which claims priority to U.S. Utility patent application Ser. No. 17/164,690 filed on Feb. 1, 2021, now U.S. Pat. No. 11,398,927, and entitled “SYSTEMS AND METHODS FOR SUBSCRIPTION BASED SELECTIVE EVPN ROUTE DISTRIBUTION” all of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

The present disclosure relates generally to communication networks, and more specifically to systems and methods for subscription based selective EVPN route distribution.

BACKGROUND

Traditionally, a route reflector in an Ethernet Virtual Private Network (EVPN) distributes route updates to all fabric provider edge nodes in an EVPN instance (EVI) in random order. A single EVPN host route update from a local provider edge node results in a Border Gateway Protocol (BGP) message from the route reflector for each of the remote provider edge nodes. However, only a handful of provider edge nodes may have active flows to a given host and therefore need the updated host route for data path convergence.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example system for subscription based selective EVPN route distribution;

FIG. 2 illustrates an example method for subscription based selective EVPN route distribution; and

FIG. 3 illustrates an example computer system that may be used by the systems and methods described herein.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

According to an embodiment, a route reflector includes one or more processors and one or more computer-readable non-transitory storage media coupled to the one or more processors and including instructions that, when executed by the one or more processors, cause the route reflector to perform operations. The operations include receiving a subscription request from a first provider edge node in a network, generating a subscription policy for the first provider edge node, and receiving a first EVPN Type 2 Route from a second provider edge node in the network. The first EVPN Type 2 Route includes at least one address of a customer edge node. The operations also include assigning a sequence number to the first EVPN Type 2 Route and communicating the first EVPN Type 2 Route with the sequence number to the first provider edge node. The operations further include receiving a second EVPN Type 2 Route from a third provider edge node. The second EVPN

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Type 2 Route includes the at least one address of the customer edge node. The operations further include generating an updated sequence number in response to receiving the second EVPN Type 2 Route from the third provider edge node and communicating the second EVPN Type 2 Route with the updated sequence number to the first provider edge node and the second provider node.

In certain embodiments, the customer edge node is locally attached to the second provider edge node when the route reflector receives the first EVPN Type 2 Route from the second provider node and the customer edge node is locally attached to the third provider edge node when the route reflector receives the second EVPN Type 2 Route from the third provider node.

In some embodiments, the subscription request includes an outbound route filter (ORF). The ORF may include an Address Family Identifier (AFI)/Subsequent Address Family Identifier (SAFI) component that is set to EVPN. The ORF may define a Route Type 2 ORF type as a sub-type-length-value (sub-TLV). The sub-TLV indicates one of the following Route Type 2 subscription types: Media Access Control (MAC) only; Internet Protocol version 4 (IPv4) only; Internet Protocol version 6 (IPv6) only; MAC+IPv4; or MAC+IPv6.

In certain embodiments, the route reflector generates the subscription policy in accordance with one of the following subscription implementation options: an exact match subscription database (DB) per update group (UG); a subscription bitmask per Route Type 2; or a Transmit List (TXLIST) per UG. In some embodiments, the first provider edge node is a border provider edge node and the customer edge node is a host. In certain embodiments, the at least one address of the customer edge node includes at least one of a MAC address and an Internet Protocol (IP) address. The network may be an EVPN Integrated Routing and Bridging (IRB) network.

According to another embodiment, a method includes receiving, by a route reflector, a subscription request from a first provider edge node in a network and generating, by the route reflector, a subscription policy for the first provider edge node. The method also includes receiving, by the route reflector, a first EVPN Type 2 Route from a second provider edge node in the network. The first EVPN Type 2 Route includes at least one address of a customer edge node. The method also includes assigning, by the route reflector, a sequence number to the first EVPN Type 2 Route and communicating, by the route reflector, the first EVPN Type 2 Route with the sequence number to the first provider edge node. The method further includes receiving, by the route reflector, a second EVPN Type 2 Route from a third provider edge node. The second EVPN Type 2 Route comprises the at least one address of the customer edge node. The method also includes generating, by the route reflector, an updated sequence number in response to receiving the second EVPN Type 2 Route from the third provider edge node and communicating, by the route reflector, the second EVPN Type 2 Route with the updated sequence number to the first provider edge node and the second provider node.

According to yet another embodiment, one or more computer-readable non-transitory storage media embody instructions that, when executed by a processor, cause the processor to perform operations. The operations include receiving a subscription request from a first provider edge node in a network, generating a subscription policy for the first provider edge node, and receiving a first EVPN Type 2 Route from a second provider edge node in the network. The first EVPN Type 2 Route includes at least one address of a

customer edge node. The operations also include assigning a sequence number to the first EVPN Type 2 Route, communicating the first EVPN Type 2 Route with the sequence number to the first provider edge node, and receiving a second EVPN Type 2 Route from a third provider edge node. The second EVPN Type 2 Route includes the at least one address of the customer edge node. The operations further include generating an updated sequence number in response to receiving the second EVPN Type 2 Route from the third provider edge node and communicating the second EVPN Type 2 Route with the updated sequence number to the first provider edge node and the second provider node.

Technical advantages of certain embodiments of this disclosure may include one or more of the following. The systems and methods for subscription based selective EVPN route distribution described herein eliminate $O(N)$ BGP messaging (one for each of the N remote provider edge nodes) for a single host route update. Messaging delay to ingress provider edge nodes with affected active flows that results from $O(N)$ messaging at a route reflector is minimized, resulting in faster data plane convergence. Since a Route Type 2 subscription model is utilized by this disclosure (rather than an on-demand pull model), unnecessary route update processing at provider edge nodes that do not have active flows is eliminated. IP and/or MAC host route scale across all provider edge nodes is minimized. $O(N)$ BGP messaging for each Route Type 2 update is eliminated at the route reflector to minimize update delay to provider edges with active flows using this Route Type 2 subscription model. BGP message processing is minimized across all provider edge nodes in the overlay. The systems and methods for subscription based selective EVPN route distribution described herein are fully backwards compatible with provider edges that do not participate in the subscription based Route Type 2 distribution. $O(1)$ BGP messages are used by certain systems and methods described herein rather than $O(N)$ BGP messages sent without route subscription policy. The $O(N)$ evaluation trade-off to evaluate the subscription policy is insignificant as compared to the $O(N)$ messaging that is eliminated.

Other technical advantages will be readily apparent to one skilled in the art from the following figures, descriptions, and claims. Moreover, while specific advantages have been enumerated above, various embodiments may include all, some, or none of the enumerated advantages.

EXAMPLE EMBODIMENTS

This disclosure describes systems and methods for subscription based selective EVPN route distribution. An EVPN route reflector traditionally distributes EVPN Route Type 2 updates to all fabric provider edge nodes in an EVI in random order. A single EVPN host route update from a local provider edge results in $O(N)$ BGP messages from the route reflector, one for each of the N remote provider edge nodes. In reality, only a handful of provider edge nodes may have active flows to a given host, and as such need the updated host route for data path convergence. Depending on the fabric provider edge scale and random update ordering on the route reflector, an $O(N)$ messaging at the route reflector may result in unwanted messaging delay to ingress provider edge nodes that actually have active flows to the host, resulting in significantly higher convergence time.

The subscription based conversational control plane procedure described herein allows an EVPN route reflector to reflect an EVPN host route to only the remote nodes that have active bridged or routed flows to the host and therefore

subscribe to the route. Certain systems and methods described herein assign a centralized sequence number at the route reflector to provide for EVPN mobility. EVPN mobility procedures, when used with subscription based selective host route distribution, allow fast roaming capability in an EVPN network by minimizing BGP Route Type 2 update generation time from the route reflector to the ingress node following a host move.

FIG. 1 illustrates an example system **100** for subscription based selective EVPN route distribution. System **100** or portions thereof may be associated with an entity, which may include any entity, such as a business or company that utilizes subscription based selective EVPN route distribution. The components of system **100** may include any suitable combination of hardware, firmware, and software. For example, the components of system **100** may use one or more elements of the computer system of FIG. 3. System **100** of FIG. 1 includes a network **110**, a route reflector **120**, a customer edge device **130**, and network nodes **140**.

Network **110** of system **100** is any type of network that facilitates communication between components of system **100**. In the illustrated embodiment of FIG. 1, network **110** is an IRB EVPN. Network **110** may connect one or more components of system **100**. One or more portions of network **110** may include an ad-hoc network, an intranet, an extranet, a VPN, an EVPN (e.g., an IRB EVPN), a local area network (LAN), a wireless LAN (WLAN), a virtual LAN (VLAN), a wide area network (WAN), a wireless WAN (WWAN), an SD-WAN, a metropolitan area network (MAN), a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a cellular telephone network, a Digital Subscriber Line (DSL), a 3G network, a 4G network, a 5G network, a Long Term Evolution (LTE) network, a combination of two or more of these, or other suitable types of networks. Network **110** may include one or more different types of networks. Network **110** may be any communications network, such as a private network, a public network, a connection through Internet, a mobile network, a WI-FI network, etc. One or more components of system **100** may communicate over network **110**. Network **110** may include a core network (e.g., the Internet), an access network of a service provider, an Internet service provider (ISP) network, and the like.

Route reflector **120** of system **100** represents any suitable network component (e.g., a physical or virtual router) that can advertise routes to nodes **140** of network **110**. In certain embodiments, route reflector **120** is a BGP route reflector. Route reflector **120** may receive routing updates from one or more nodes **140** of network **110**. In some embodiments, route reflector **120** reflects EVPN prefixes to nodes **140** (e.g., provider edge nodes) such that nodes **140**. In certain embodiments, route reflector **120** may be an edge router (e.g., a subscriber edge router, a label edge router, etc.), a BGP router, a wired router, a wireless router, a virtual router, a core router, a gateway, a combination thereof, or any other suitable type of router. In certain embodiments, route reflector **120** may receive, create, process, store, and/or send traffic to other components within network **110**. Route reflector **120** may be controlled by one or more entities (e.g., a network service provider). Route reflector **120** may be located in one or more data centers, remote branches, or any other suitable location.

Customer edge device **130** of system **100** is a device that is identified by a MAC and/or IP address. In certain embodiments, customer edge device **130** may be a host, a router, a switch, and the like. Customer edge device **130** may send and/or receive data within network **110**. In some embodi-

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ments, customer edge device **130** connects to one or more nodes **140** of network **110**. For example, as illustrated in FIG. 1, customer edge device **130** may connect to node **140a** and/or node **140b** of network **110**.

Nodes **140** of network **110** are connection points within network **110** that receive, create, store and/or send data along a path. Nodes **140** may include one or more redistribution points that recognize, process, and forward data to other nodes **140** of network **110**. Nodes **140** may include virtual and/or physical edge nodes. In certain embodiments, one or more nodes **140** include data communications equipment such as routers, servers, printers, workstations, switches, bridges, modems, hubs, and the like. Nodes **140** may include provider edge nodes, customer edge nodes, border nodes, etc.

In the illustrated embodiment of FIG. 1, nodes **140** include node **140a**, node **140b**, and node **140c**. In certain embodiments, route reflector **120**, node **140a**, node **140b**, and node **140c** are configured to send and/or receive EVPN advertisement routes. For example, as illustrated in FIG. 1, node **140a** communicates EVPN advertisement route **170** to route reflector **120**, and node **140b** communicates EVPN advertisement route **180** to route reflector **120**. As another example, as illustrated in FIG. 1, route reflector **120** communicates EVPN advertisement route **170** to node **140c**, and route reflector **120** communicates EVPN advertisement route **180** to node **140a** and node **140c**. EVPN advertisement routes **170** and **180** may include BGP EVPN advertisement routes such as Type 2 routes, Type 3 routes, Type 5 routes, and the like. EVPN Type 2 routes are MAC/IP routes that may be used by nodes **140** of network **110** to advertise MAC and/or IP addresses, address reachability, Address Resolution Protocol (ARP) entry, IP/MAC binding, and/or host IP routing information to other nodes **140**. In some embodiments, nodes **140** include one or more border nodes. For example, node **140c** of FIG. 1 may be a border node that is configured as a default route across all provider edge nodes and therefore may be configured to receive all EVPN advertisement route updates (e.g., EVPN Type 2 route updates).

Flow Based Route Subscription Requirements. Across EVPN symmetric and asymmetric IRB use cases, depending on active flows via nodes **140**, nodes **140** may require one or more of the following: (1) a remote MAC route to be installed in MAC-virtual routing and forwarding (VRF) to forward bridged flows; (2) a remote IP host route to be installed in IP-VRF to forward host routed flows in a Symmetric IRB scenario; (3) a remote IP→MAC binding to be installed as an adjacency or next-hop forwarding object in an Asymmetric IRB scenario; and/or (4) a remote IP→MAC binding to be installed in ARP suppression cache.

As such, the route subscription requirements may include one or more of the following: (1) Subscription based route distribution is only required for EVPN MAC+IP and MAC only Type 2 routes. Distribution of all other EVPN route types is not affected. (2) Each node **140** that participates in subscription based route distribution does not receive any Route Type 2 update as a default policy, unless that particular node **140** overrides the default policy with a Route Type 2 subscription. (3) The Route Type 2 subscription mechanism between nodes **140** and route reflector **120** must support route subscription from nodes **140** to route reflector **120**. The subscription mechanism may be identified by: (a) “a host IP and an import Route Target (RT) set” that allows MAC+IP Route Type 2 updates from route reflector **120** to nodes **140** that exactly match the specified IP and at least one of the RT(s) in the RT set; (b) “a host IP, MAC, and an

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import RT set” that allows MAC+IP Route Type 2 updates from route reflector **120** to nodes **140** that exactly match the specified IP, MAC, and at least one of the RT(s) in the RT set; (c) “a host MAC and an import RT set” that allows MAC+IP and MAC Route Type 2 updates that exactly match the specified MAC, and any of the RT(s) in the RT list; and/or (d) “a wild card” that allows a border node (e.g., node **140c**) to subscribe for all Route Type 2 updates by default. An explicit wild card registration/un-registration may allow border role transitions to be handled without a BGP session flap. (4) Subscription is implicit for nodes **140** that originate a local Route Type 2 path. For example, an originator of a Route Type 2 path is implicitly subscribed for the Route Type 2 updates.

EVPN Route Type 2 Subscription Signaling. The BGP ORF framework defined in RFC 5291, which is incorporated herein by reference, is used as a foundation and further extended to signal EVPN Route Type 2 subscriptions from nodes **140** to route reflector **120**. A Route Type 2 subscription request is signaled as an outbound route filter policy from nodes **140** to route reflector **120**. AFI/SAFI is set to EVPN. A new ORF type is defined for EVPN Route Type 2. In certain embodiments, ORF entry for Route Type 2 ORF type is defined as a sub-TLV, where a sub-type indicates one of the following Route Type 2 subscription types: (1) MAC only; contents are MAC address and RT(s). Length is MAC address length (6 bytes)+RT TLV length; (2) IPv4 only; contents are IPv4 address and RT(s). Length is IPv4 address length (4 bytes)+RT TLV length; (3) IPv6 only; contents are IPv6 address and RT(s). Length is IPv6 address length (16 bytes)+RT TLV length; (4) MAC+IPv4; contents are MAC+IPv4 and RT(s). Length is MAC address length (6 bytes)+IPv4 address length (4 bytes)+RT TLV length; and (5) MAC+IPv6; contents are MAC+IPv6 and RT(s). Length is MAC address length (6 bytes)+IPv6 address length (16 bytes)+RT TLV length. These Route Type 2 subscription types cover IRB use cases across both symmetric and asymmetric IRB architectures.

Subscription Based Route Type 2 Update Generation. In certain embodiments, each node **140** that signals a Route Type 2 subscription policy results in a separate UG at route reflector **120**. ‘N’ nodes **140** in a fabric may result in ‘N’ UGs. As a result, route reflector **120** may evaluate a Route Type 2 update ‘N’ times, but if one node **140** has signaled a matching subscription policy, only a single BGP message is formatted and sent. While ORF is used as a signaling mechanism to signal Route Type 2 subscription, this does not necessarily imply using existing ORF implementation for Route Type 2 outbound policy enforcement. To further minimize O(N) evaluation tradeoff resulting from Route Type 2 subscription based update generation at route reflector **120**, more efficient alternatives are considered: (1) Existing ORF implementations; (2) Exact match subscription DB per UG; (3) Subscription bitmask per Route Type 2; and (4) TXLIST per UG. The above implementation options for subscription based Route Type 2 update generation are described as follows:

(1) Existing ORF implementations. Existing ORF implementations typically use a prefix-list for each UG to track prefix-based ORF policies. This prefix-list is internally implemented as a radix trie. For a route update, each of ‘N’ UGs is evaluated via its ORF prefix-list radix lookup. The radix lookup result determines whether the route is sent to that UG. The prefix-list/radix based ORF implementation is not efficient for host IP routes. The prefix-list/radix based ORF implementation is not suitable for exact-match applications like MAC-IP or MAC Route Type 2.

(2) Exact match subscription DB per UG. 'N' UGs are used for 'N' nodes **140**. A Route Type 2 subscription Adelson-Velsky and Landis (AVL) DB is maintained for each UG. For a Route Type 2 update, each of 'N' UGs is evaluated via a lookup in that UG's Route Type 2 subscription DB. The lookup result determines if a Route Type 2 update is sent to the UG/node **140**. As such, the evaluation cost per Route Type 2 is $O(N \log(n))$, where N =number of nodes **140**, and n =number of destinations for active flows at node **140**. Route reflector **120** may evaluate a Route Type 2 update 'N' times, but if only one node **140** has signaled a matching subscription policy, only a single BGP message is formatted and sent. The subscription DB size 'n' is minimal since destinations for active flows are tracked at nodes **140**. Most $\log(n)$ lookups result in no match and no update generation.

(3) Subscription bitmask per Route Type 2. 'N' UGs are used for 'N' nodes **140**. A subscription bitmask is maintained per-route to track UGs subscribed for that route. For a Route Type 2 update, each UG is evaluated via the route's bitmask check for that UG. The bitmask check result determines if a Route Type 2 update is sent to that UG/node **140**. As such, the evaluation cost per Route Type 2 is $O(N)$, where N =number of nodes **140**. Route reflector **120** may evaluate a Route Type 2 update 'N' times, but if only one node **140** has signaled a matching subscription policy, only a single BGP message is formatted and sent. In certain embodiments, a subscription DB per-UG/node **140** may still be required to efficiently update affected route subscriptions on events like node **140** session down.

(4) TXLIST per UG. 'N' UGs are used for 'N' nodes **140**. A TXLIST is maintained per UG/node **140**. A Route Type 2 subscription by node **140** results in Route Type 2 added to the UG TXLIST. ROUTE TYPE 2 maintains, inline, a dynamic list of UG TXLIST link-points. On a Route Type 2 update, each TXLIST in this dynamic list is processed to schedule a Route Type 2 update to that UG/node **140** (typically by moving it to the back of the TXLIST). UGs that have not subscribed to this Route Type 2 are not evaluated. While memory overhead of multiple TXLIST link-points per-route is required, this option has no evaluation tradeoff.

Above options (2), (3), or (4), or another implementation found more suitable, may be selected for subscription based Route Type 2 update generation, depending on performance/complexity tradeoffs at implementation time. Nodes **140** that do not participate in Route Type 2 subscription capability continue to receive all Route Type 2 updates as per RFC 5291, which ensures full backwards compatibility.

In the illustrated embodiment of FIG. 1, node **140c** (e.g., a border node) communicates (see notation **150**) a subscription request **122** to route reflector **120**. In certain embodiments, subscription request **122** is an EVPN Route Type 2 subscription request that is signaled as an outbound route filter policy from node **140c** to route reflector **120**. AFI/SAFI may be set to EVPN, and a new ORF type may be defined for EVPN Route Type 2. In response to receiving subscription request **122**, route reflector **140** generates a subscription policy **124** for node **140c**. In certain embodiments, subscription policy **124** is an EVPN Route Type 2 subscription policy. Route reflector **140** may generate a separate UG for each subscription policy **124**.

Customer edge device **130** of FIG. 1 (e.g., a host) connects to node **140a** of network **110**. In response to node **140a** detecting a locally attached MAC and/or IP address for customer edge device **130**, node **140a** communicates (see notation **152**) an EVPN advertisement route **170**. In certain embodiments, EVPN advertisement route **170** is an EVPN

Type 2 Route. Upon receiving EVPN advertisement route **170** from node **140a**, route reflector **120** communicates (see notation **154**) EVPN advertisement route **170** (e.g., an EVPN Type 2 route) to subscriber node **140c**. Subscriber node **140c** now has EVPN advertisement route **170** for customer edge device **130** in MAC and/or IP VRF, depending on the nature of the flow. Node **140c** communicates network traffic along path **156** in accordance with EVPN advertisement route **170**.

Sequence Number Assignment Procedure. In certain embodiments, customer edge device **130** may move between nodes **140**. For example, customer edge device **130** may disconnect from node **140a** and connect to node **140b**. Since Route Type 2 updates are not distributed to all nodes **140**, EVPN mobility procedures as defined in RFC 7432, which is incorporated herein by reference, no longer work. To address this issue, a centralized Route Type 2 sequence number assignment procedure may be implemented at route reflector **130** when subscription based host route distribution is deployed. This centralized sequence number assignment allows existing EVPN mobility procedures on nodes **140** to function as defined in RFC 7432.

In the illustrated embodiment of FIG. 1, node **140a** discovers customer edge device **130** and learns a local MAC and/or IP address from customer edge device **130**. Node **140a** assigns the local MAC and/or IP a sequence number **172** as per RFC 7432. If this is the first mobility event for the local MAC and/or IP address of customer edge device **130**, EVPN advertisement route **170** communicated from node **140a** to route reflector **120** does not carry a sequence number via a MAC mobility extended community attribute. As such, the value of sequence number **172** for the first mobility event for the local MAC and/or IP address of customer edge device **130** is assumed to be 0.

Route reflector **120**, on receiving EVPN advertisement route **170**, reflects (see notation **154**) EVPN advertisement route **170** with sequence number **172** (e.g., 0) to node **140c** that has subscribed to this Route Type 2. Customer edge device **130** then moves (see notation **158**) from node **140a** to node **140b**. Node **140b** communicates (see notation **160**) EVPN advertisement route **180** to route reflector **120**. Upon receiving EVPN advertisement route **180** from node **140b**, route reflector **120** compares EVPN advertisement route **180** against any pre-existing best path (e.g., EVPN advertisement route **170**) with a different Ethernet Segment Identifier (ESI). Based on the comparison, route reflector **120** updates sequence number **172** to generate an updated sequence number **182**. For example, route reflector **120** may update sequence number **172** by a value of one (e.g., from a value of 0 to a value of 1). Route reflector **120** communicates (see notation **162**) updated EVPN advertisement route **180** with updated sequence number **182** to node **140c** since node **140c** subscribed to Route Type 2. Node **140c** then communicates network traffic along updated path **164** in accordance with updated EVPN advertisement route **180**. Originating node **140a** for the previous best path is implicitly subscribed and hence also receives EVPN advertisement route **180** with updated sequence number **182**, which allows node **140a** to trigger cleanup/duplicate detection procedures for its local path. This sequence number method ensures backwards compatibility with any node **140** that is not participating in subscription-based route distribution.

Sequence Number Assignment Across Paths with Different Route Distinguishers. An EVI requires a Route Distinguisher (RD) that is unique per MAC-VRF and one or more globally unique RTs. In a typical topology where each node **140** uses a different RD, route reflector **120** computes the

best path only across all Route Type 2 paths received with a common RD. However, centralized sequence number assignment logic on route reflector **120** for this case requires sequence number comparison across paths with Route Type 2 having different RDs, and hence requires best path computation across all RDs. Centralized sequence number computation on route reflector **120** is enhanced to mimic route server “like” function with local MAC VRF import to accomplish sequence number comparison across Route Type 2 paths with different RDs. However, this MAC VRF import within BGP is only for the purpose of sequence number assignment and is not intended to allow any data path through router reflector **120**. This MAC VRF import may be limited to BGP and does not result in the route being downloaded to routing information base (RIB)/forwarding information base (FIB) MAC VRF.

In certain embodiments, route reflector **120** is configured with MAC VRFs in a given fabric and corresponding fabric RTs to perform local MAC VRF import in a mode that does not install any MAC VRF routes in RIB. RT may be auto-derived as autonomous system (AS): Virtual Network Instance (VNI), which results in the same RT across all nodes **140** and route reflector **120** in the fabric AS. In the case of external BGP (eBGP), the AS number may be rewritten to local AS in RT. In some embodiments, a unique RD is derived using Router-ID: AS-Num. In a uniformly orchestrated fabric with the same RT across all nodes **140**, MAC VRFs, corresponding RD, RT, and import policy may be auto-created on route reflector **120** for the purpose of sequence number assignment.

Although FIG. 1 illustrates a particular arrangement of network **110**, route reflector **120**, subscription request **122**, subscription policy **124**, customer edge device **130**, nodes **140**, EVPN advertisement route **170**, sequence number **172**, EVPN advertisement route **180**, and updated sequence number **182**, this disclosure contemplates any suitable arrangement of network **110**, route reflector **120**, subscription request **122**, subscription policy **124**, customer edge device **130**, nodes **140**, EVPN advertisement route **170**, sequence number **172**, EVPN advertisement route **180**, and updated sequence number **182**.

Although FIG. 1 illustrates a particular number of networks **110**, route reflectors **120**, subscription requests **122**, subscription policies **124**, customer edge devices **130**, nodes **140**, EVPN advertisement routes **170**, sequence numbers **172**, EVPN advertisement routes **180**, and updated sequence numbers **182**, this disclosure contemplates any suitable number of networks **110**, route reflectors **120**, subscription requests **122**, subscription policies **124**, customer edge devices **130**, nodes **140**, EVPN advertisement routes **170**, sequence numbers **172**, EVPN advertisement routes **180**, and updated sequence numbers **182**. For example, system **100** may include more than one route reflector **120** and/or more or less than three nodes **140**.

FIG. 2 illustrates an example method **200** for subscription based selective EVPN route distribution. Method **200** begins at step **205**. At step **210**, a network component (e.g., route reflector **120** of FIG. 1) receives a subscription request (e.g., subscription request **122** of FIG. 1) from a first provider edge node (e.g., node **140c** of FIG. 1) in a network (e.g., network **110** of FIG. 1). In certain embodiments, the network is an EVPN IRB network. The first provider edge node may be a border provider edge node. In some embodiments, the subscription request includes an ORF, which may include an AFI/SAFI component that is set to EVPN. In some embodiments, the ORF defines a Route Type 2 ORF type as a

sub-TLV. The sub-TLV indicates one of the following Route Type 2 subscription types: MAC only; IPv4 only; IPv6 only; MAC+IPv4; or MAC+IPv6.

Method **200** then moves from step **210** to step **215**, where the network component generates a subscription policy (e.g., subscription policy **124** of FIG. 1) for the first provider edge node. The subscription policy may be generated in accordance with one of the following subscription implementation options: an exact match subscription DB per UG; a subscription bitmask per Route Type 2; or a TXLIST per UG. Method **200** then moves from step **215** to step **220**.

At step **220** of method **200**, the network component receives a first EVPN Type 2 Route (e.g., EVPN advertisement route **170** of FIG. 1) from a second provider edge node (e.g., node **140a** of FIG. 1) in the network. The first EVPN Type 2 Route includes a MAC and/or IP address of a customer edge node (e.g., customer edge device **130** of FIG. 1). In certain embodiments, the customer edge node is a host. The customer edge node is locally attached to the second provider edge node when the network component receives the first EVPN Type 2 Route from the second provider node.

Method **200** then moves from step **220** to step **225**, where the network component assigns a sequence number (e.g., sequence number **172** of FIG. 1) to the first EVPN Type 2 Route. If this is the first mobility event for the local MAC and/or IP address of the customer edge device, the first EVPN Type 2 Route communicated from the second provider edge node to the route reflector will not carry a sequence number via a MAC mobility extended community attribute. As such, the value of the sequence number for the first mobility event for the local MAC and/or IP address of the customer edge device is assumed to be 0. Method **200** then moves from step **225** to step **230**.

At step **230** of method **200**, the network component communicates the first EVPN Type 2 Route with the sequence number to the first provider edge node since the first provider edge subscribed to the EVPN Type 2 Route. Method **200** then moves from step **230** to step **235**, where method **200** determines whether the customer edge node moved from the second provider edge node to a third provider edge node (e.g., node **140b** of FIG. 1). If the customer edge node is still attached to the second provider edge node, method **200** advances from step **235** to step **255**, where method **200** ends.

If, at step **235**, the customer edge node moved from the second provider edge node and attached itself to the third provider edge node, method **200** moves from step **235** to step **240**, where the network component receives a second EVPN Type 2 Route (e.g., EVPN Type 2 Route **180**) from the third provider edge node. The second EVPN Type 2 Route includes the MAC and/or IP address of the customer edge node. The customer edge node is locally attached to the third provider edge node when the network component receives the second EVPN Type 2 Route from the third provider node. Upon receiving the second EVPN Type 2 Route from the third provider edge node, the network component compares the second EVPN Type 2 Route against any pre-existing best path (e.g., the first EVPN Type 2 Route) with a different ESI. Method **200** then moves from step **240** to step **245**.

At step **245** of method **200**, the network component generates an updated sequence number (e.g., sequence number **182** of FIG. 1) in response to receiving the second EVPN Type 2 Route from the third provider edge node. For example, based on the comparison of the second EVPN Type 2 Route against the first EVPN Type 2 Route, the

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network component may update the sequence number by a value of one (e.g., from a value of 0 to a value of 1). Method 200 then moves from step 245 to step 250, where the network component communicates the second EVPN Type 2 Route with the updated sequence number to the first provider edge node since the first provider edge node subscribed to the EVPN Type 2 Route. The network component also communicates the second EVPN Type 2 Route with the updated sequence number to the second provider edge node since the second provider edge node is the originating node for the previous best path. The originating node is implicitly subscribed to the EVPN Type 2 Route 180, which allows the originating node to trigger cleanup/duplicate detection procedures for its local path. Method 200 then moves from step 250 to step 255, where method 200 ends.

Although this disclosure describes and illustrates particular steps of the method of FIG. 2 as occurring in a particular order, this disclosure contemplates any suitable steps of the method of FIG. 2 occurring in any suitable order. For example, method 200 may include a step where the customer edge node moves from the third provider edge node to a fourth provider edge node. Moreover, although this disclosure describes and illustrates an example method for subscription based selective EVPN route distribution including the particular steps of the method of FIG. 2, this disclosure contemplates any suitable method for subscription based selective route distribution including any suitable steps, which may include all, some, or none of the steps of the method of FIG. 2, where appropriate. Furthermore, although this disclosure describes and illustrates particular components, devices, or systems carrying out particular steps of the method of FIG. 2, this disclosure contemplates any suitable combination of any suitable components, devices, or systems carrying out any suitable steps of the method of FIG. 2.

FIG. 3 illustrates an example computer system 300. In particular embodiments, one or more computer systems 300 perform one or more steps of one or more methods described or illustrated herein. In particular embodiments, one or more computer systems 300 provide functionality described or illustrated herein. In particular embodiments, software running on one or more computer systems 300 performs one or more steps of one or more methods described or illustrated herein or provides functionality described or illustrated herein. Particular embodiments include one or more portions of one or more computer systems 300. Herein, reference to a computer system may encompass a computing device, and vice versa, where appropriate. Moreover, reference to a computer system may encompass one or more computer systems, where appropriate.

This disclosure contemplates any suitable number of computer systems 300. This disclosure contemplates computer system 300 taking any suitable physical form. As example and not by way of limitation, computer system 300 may be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC) (such as, for example, a computer-on-module (COM) or system-on-module (SOM)), a desktop computer system, a laptop or notebook computer system, an interactive kiosk, a mainframe, a mesh of computer systems, a mobile telephone, a personal digital assistant (PDA), a server, a tablet computer system, an augmented/virtual reality device, or a combination of two or more of these. Where appropriate, computer system 300 may include one or more computer systems 300; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or

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more networks. Where appropriate, one or more computer systems 300 may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example and not by way of limitation, one or more computer systems 300 may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems 300 may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

In particular embodiments, computer system 300 includes a processor 302, memory 304, storage 306, an input/output (I/O) interface 308, a communication interface 310, and a bus 312. Although this disclosure describes and illustrates a particular computer system having a particular number of particular components in a particular arrangement, this disclosure contemplates any suitable computer system having any suitable number of any suitable components in any suitable arrangement.

In particular embodiments, processor 302 includes hardware for executing instructions, such as those making up a computer program. As an example and not by way of limitation, to execute instructions, processor 302 may retrieve (or fetch) the instructions from an internal register, an internal cache, memory 304, or storage 306; decode and execute them; and then write one or more results to an internal register, an internal cache, memory 304, or storage 306. In particular embodiments, processor 302 may include one or more internal caches for data, instructions, or addresses. This disclosure contemplates processor 302 including any suitable number of any suitable internal caches, where appropriate. As an example and not by way of limitation, processor 302 may include one or more instruction caches, one or more data caches, and one or more translation lookaside buffers (TLBs). Instructions in the instruction caches may be copies of instructions in memory 304 or storage 306, and the instruction caches may speed up retrieval of those instructions by processor 302. Data in the data caches may be copies of data in memory 304 or storage 306 for instructions executing at processor 302 to operate on; the results of previous instructions executed at processor 302 for access by subsequent instructions executing at processor 302 or for writing to memory 304 or storage 306; or other suitable data. The data caches may speed up read or write operations by processor 302. The TLBs may speed up virtual-address translation for processor 302. In particular embodiments, processor 302 may include one or more internal registers for data, instructions, or addresses. This disclosure contemplates processor 302 including any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor 302 may include one or more arithmetic logic units (ALUs); be a multi-core processor; or include one or more processors 302. Although this disclosure describes and illustrates a particular processor, this disclosure contemplates any suitable processor.

In particular embodiments, memory 304 includes main memory for storing instructions for processor 302 to execute or data for processor 302 to operate on. As an example and not by way of limitation, computer system 300 may load instructions from storage 306 or another source (such as, for example, another computer system 300) to memory 304. Processor 302 may then load the instructions from memory 304 to an internal register or internal cache. To execute the instructions, processor 302 may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor 302

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may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor **302** may then write one or more of those results to memory **304**. In particular embodiments, processor **302** executes only instructions in one or more internal registers or internal caches or in memory **304** (as opposed to storage **306** or elsewhere) and operates only on data in one or more internal registers or internal caches or in memory **304** (as opposed to storage **306** or elsewhere). One or more memory buses (which may each include an address bus and a data bus) may couple processor **302** to memory **304**. Bus **312** may include one or more memory buses, as described below. In particular embodiments, one or more memory management units (MMUs) reside between processor **302** and memory **304** and facilitate accesses to memory **304** requested by processor **302**. In particular embodiments, memory **304** includes random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Moreover, where appropriate, this RAM may be single-ported or multi-ported RAM. This disclosure contemplates any suitable RAM. Memory **304** may include one or more memories **304**, where appropriate. Although this disclosure describes and illustrates particular memory, this disclosure contemplates any suitable memory.

In particular embodiments, storage **306** includes mass storage for data or instructions. As an example and not by way of limitation, storage **306** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **306** may include removable or non-removable (or fixed) media, where appropriate. Storage **306** may be internal or external to computer system **300**, where appropriate. In particular embodiments, storage **306** is non-volatile, solid-state memory. In particular embodiments, storage **306** includes read-only memory (ROM). Where appropriate, this ROM may be mask-programmed ROM, programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), electrically alterable ROM (EAROM), or flash memory or a combination of two or more of these. This disclosure contemplates mass storage **306** taking any suitable physical form. Storage **306** may include one or more storage control units facilitating communication between processor **302** and storage **306**, where appropriate. Where appropriate, storage **306** may include one or more storages **306**. Although this disclosure describes and illustrates particular storage, this disclosure contemplates any suitable storage.

In particular embodiments, I/O interface **308** includes hardware, software, or both, providing one or more interfaces for communication between computer system **300** and one or more I/O devices. Computer system **300** may include one or more of these I/O devices, where appropriate. One or more of these I/O devices may enable communication between a person and computer system **300**. As an example and not by way of limitation, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, still camera, stylus, tablet, touch screen, trackball, video camera, another suitable I/O device or a combination of two or more of these. An I/O device may include one or more sensors. This disclosure contemplates any suitable I/O devices and any suitable I/O interfaces **308** for them. Where appropriate, I/O interface **308** may include one or more device or software drivers enabling processor **302** to drive one or more of these I/O devices. I/O interface **308** may include one or more I/O interfaces **308**, where

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appropriate. Although this disclosure describes and illustrates a particular I/O interface, this disclosure contemplates any suitable I/O interface.

In particular embodiments, communication interface **310** includes hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **300** and one or more other computer systems **300** or one or more networks. As an example and not by way of limitation, communication interface **310** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC) or wireless adapter for communicating with a wireless network, such as a WI-FI network. This disclosure contemplates any suitable network and any suitable communication interface **310** for it. As an example and not by way of limitation, computer system **300** may communicate with an ad hoc network, a personal area network (PAN), a LAN, a WAN, a MAN, or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **300** may communicate with a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a WI-MAX network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network, a 3G network, a 4G network, a 5G network, an LTE network, or other suitable wireless network or a combination of two or more of these. Computer system **300** may include any suitable communication interface **310** for any of these networks, where appropriate. Communication interface **310** may include one or more communication interfaces **310**, where appropriate. Although this disclosure describes and illustrates a particular communication interface, this disclosure contemplates any suitable communication interface.

In particular embodiments, bus **312** includes hardware, software, or both coupling components of computer system **300** to each other. As an example and not by way of limitation, bus **312** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCIe) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or another suitable bus or a combination of two or more of these. Bus **312** may include one or more buses **312**, where appropriate. Although this disclosure describes and illustrates a particular bus, this disclosure contemplates any suitable bus or interconnect.

Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such as, for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs, optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, solid-state drives (SSDs), RAM-drives, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable

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non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, feature, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Additionally, although this disclosure describes or illustrates particular embodiments as providing particular advantages, particular embodiments may provide none, some, or all of these advantages.

What is claimed is:

1. A method, comprising:
 - receiving a subscription request from a first Border Gateway Protocol (BGP) node in a network, the subscription request comprising a first BGP route advertisement, wherein the first BGP route advertisement contains a route filter for an Address Family Identifier (AFI)/Subsequent Address Family Identifier (SAFI);
 - generating a subscription policy in accordance with the route filter contained in the subscription request;
 - receiving a second BGP route advertisement from a second BGP node in the network; and
 - communicating routing information in the second BGP route advertisement to the first BGP node based on the subscription policy.
2. The method of claim 1, wherein the subscription request is an Ethernet Virtual Private Network (EVPN) subscription request.
3. The method of claim 2, wherein the subscription request is an EVPN Type 2 subscription request.
4. The method of claim 1, wherein the subscription request results in a modification of an Outbound Route Filter (ORF).
5. The method of claim 1, wherein the subscription request is based on an active flow associated with the first BGP node.
6. The method of claim 1, wherein the subscription request and subsequent routing information traverse at least one intermediate node.

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7. The method of claim 1, wherein the first BGP node is a provider edge node.

8. The method of claim 1, wherein the subscription request is received by a route reflector.

9. An apparatus, comprising:

one or more processors; and

one or more computer-readable non-transitory storage media coupled to the one or more processors and comprising instructions that, when executed by the one or more processors, cause the apparatus to perform operations comprising:

receiving a subscription request from a first Border Gateway Protocol (BGP) node in a network, the subscription request comprising a first BGP route advertisement, wherein the first BGP route advertisement contains a route filter for an Address Family Identifier (AFI)/Subsequent Address Family Identifier (SAFI);

generating a subscription policy in accordance with the route filter contained in the subscription request;

receiving a second BGP route advertisement from a second BGP node in the network; and

communicating routing information in the second BGP route advertisement to the first BGP node based on the subscription policy.

10. The apparatus of claim 9, wherein the subscription request is an Ethernet Virtual Private Network (EVPN) subscription request.

11. The apparatus of claim 10, wherein the subscription request is an EVPN Type 2 subscription request.

12. The apparatus of claim 9, wherein the subscription request results in a modification of an Outbound Route Filter (ORF).

13. The apparatus of claim 9, wherein the subscription request is based on an active flow associated with the first BGP node.

14. The apparatus of claim 9, wherein the subscription request and subsequent routing information traverse at least one intermediate BGP node.

15. The apparatus of claim 9, wherein the first BGP node is a provider edge node.

16. The apparatus of claim 9, wherein the subscription request is received by a route reflector.

17. One or more computer-readable non-transitory storage media embodying instructions that, when executed by a processor, cause the processor to perform operations comprising:

receiving a subscription request from a first Border Gateway Protocol (BGP) node in a network, the subscription request comprising a first BGP route advertisement, wherein the first BGP route advertisement contains a route filter for an Address Family Identifier (AFI)/Subsequent Address Family Identifier (SAFI);

generating a subscription policy in accordance with the route filter contained in the subscription request;

receiving a second BGP route advertisement from a second BGP node in the network; and

communicating routing information in the second BGP route advertisement to the first BGP node based on the subscription policy.

18. The one or more computer-readable non-transitory storage media of claim 17, wherein the subscription request is an Ethernet Virtual Private Network (EVPN) subscription request.

19. The one or more computer-readable non-transitory storage media of claim 18, wherein the subscription request is an EVPN Type 2 subscription request.

20. The one or more computer-readable non-transitory storage media of claim 17, wherein the subscription request results in a modification of an Outbound Route Filter (ORF).

21. The one or more computer-readable non-transitory storage media of claim 17, wherein the subscription request is based on an active flow associated with the first BGP node. 5

22. The one or more computer-readable non-transitory storage media of claim 17, wherein the subscription request and subsequent routing information traverse at least one intermediate BGP node. 10

23. The one or more computer-readable non-transitory storage media of claim 17, wherein the first BGP node is a provider edge node.

24. The one or more computer-readable non-transitory storage media of claim 17, wherein the subscription request is received by a route reflector. 15

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